

Façade Element Detection for Continuous Value Engineering

A YOLOv8 detector that reads building-envelope elements from photographs and rendered elevations — so a Value Engineering assistant can estimate envelope composition for a building it has **no BIM model of**.

Omar Elsayed · Group 02

5 classes

YOLOv8n · 30 epochs · 640 px

80 / 20 split

Apache-2.0

The BIM model cannot answer the question at the moment it is asked

Continuous Value Engineering needs to know what is on the façade.
Three situations put that truth **outside** the model:

- **Concept & early SD** — no modelled façade exists yet, only renders and sketch elevations.
- **Benchmarking** — there is no BIM model of a precedent or a competitor's scheme.
- **As-built divergence** — the façade was value-engineered on site. The model says one thing; the photograph says another.

Why the envelope. Typically **15–25 % of construction cost** and consistently the first package opened in a VE exercise. An early, cheap, repeatable read on envelope composition is worth more than a more accurate read on almost anything else.

SUCCESS CRITERIA

Recover **Window-to-Wall Ratio to within ± 10 %** of a human's reading of the same image, on frontal-to-30°-oblique daylight elevations of 3–10 storey buildings, in under **2 seconds** per image.

THE ANTI-USE CASE

Not a quantity take-off. Not a compliance check. Not a BIM-vs-reality verifier.
Not a people-watching system — there is **no person class, by design**.

Baseline check (Session 1). Stock COCO YOLOv8 on a façade detects no windows, no curtain wall and no balconies. It is not inaccurate — it is answering a different question. COCO has `toothbrush` ; it has no `curtain_wall` .

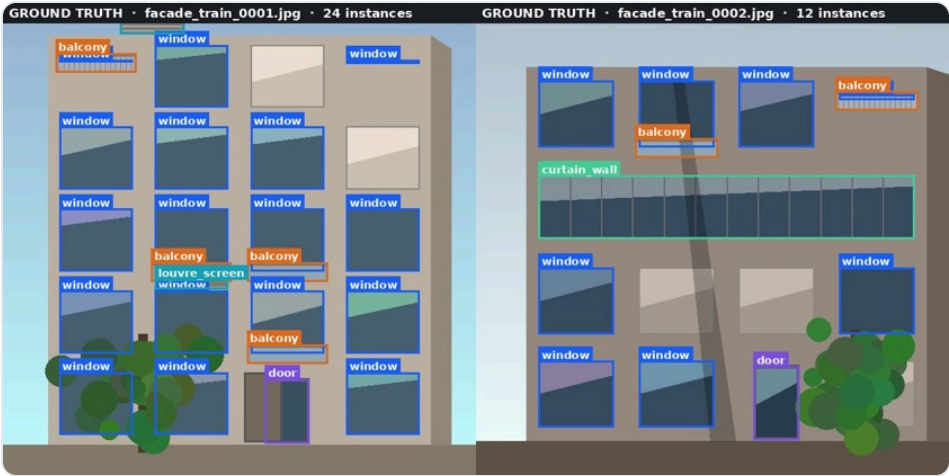
Five classes, three standing rules, one load-bearing order

ID	CLASS	THE DISCRIMINATING TEST
0	■ window	bounded by opaque material on all four sides
1	■ curtain_wall	<i>can I trace unbroken glass across it?</i>
2	■ door	at grade and enterable
3	■ balcony	occupiable depth, not a guard rail
4	■ louvre_screen	permanent, not scaffold netting

THE THREE STANDING RULES

- 1. No reflections — a mirrored window double-counts the envelope.
- 2. No neighbouring buildings — one subject per image.
- 3. Partial and occluded elements **are** labelled, visible extent only, 8 px floor.

Boxes are drawn to the **outer frame edge** — the frame is part of the purchased unit, therefore part of the cost. Class ids are positional, so the loader **refuses to train** if a dataset's order disagrees with the config.



Ground truth as applied — generated from the committed labels, not screenshotted.

SAM VERDICT

Helped on **curtain_wall, door, balcony** — large convex regions with strong edges. Failed on the **window / curtain_wall** boundary: SAM sees edges, not the contract. It segments mirror-glass spandrels beautifully — and they must not be labelled at all.

yolov8n · 30 epochs · batch 16 · imgsiz 640 · seed 42

0.975

PRECISION

0.946

RECALL

0.948

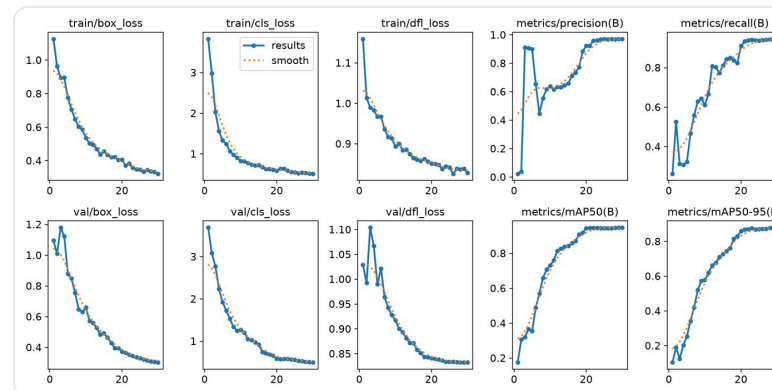
MAP50

0.879

MAP50-95 · HEADLINE

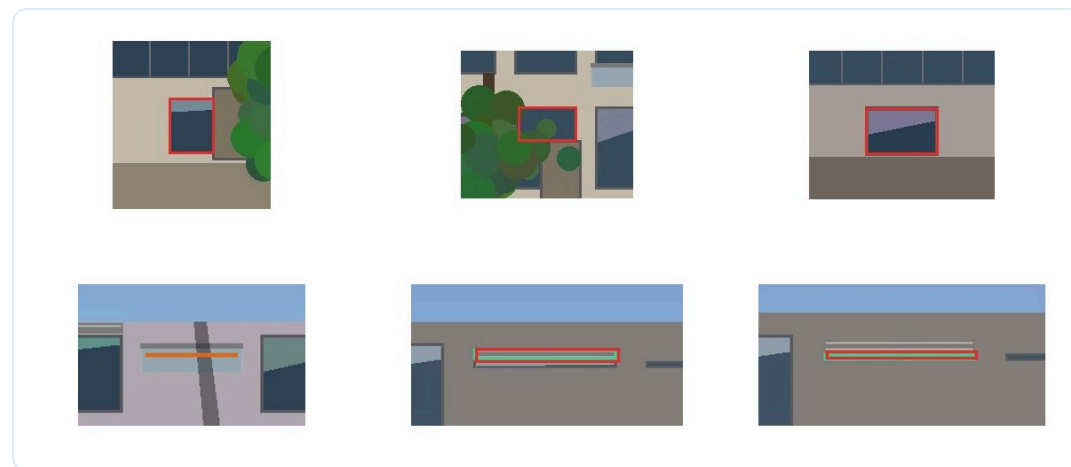
CLASS	INST.	P	R	MAP50	MAP50-95
window	316	0.980	0.951	0.985	0.951
curtain_wall	37	0.980	1.000	0.995	0.976
door	24	0.952	0.830	0.821	0.809
balcony	66	0.994	0.970	0.965	0.897
louvre_screen	48	0.971	0.979	0.971	0.764

Verification run, not a performance claim. Procedural dataset, no credentials required — it proves the pipeline executes end to end for a stranger. Synthetic façades are far easier than photographs. Expect materially lower numbers on real data.



Why mAP50-95 is the headline. The VE read-out is computed from **box area**. A box 20 % too large scores a clean hit at IoU 0.50 and produces a 20 % cost error. **mAP50** is blind to exactly the failure that costs this project money.

The metric you report decides what you think you built



■ ground truth
 ■ wrong prediction
 ■ missed instance

ERROR TYPE	COUNT	% OF ERRORS
False negative	23	56.1 %
False positive	9	22.0 %
Localization error	6	14.6 %
Class confusion	3	7.3 %

The finding. `louvre_screen` scores mAP50 **0.971** and mAP50-95 **0.764**. The model finds almost every louvre screen and boxes almost none tightly — worst case **+31 % box area**. Every one is a true positive at IoU 0.50. A project quoting mAP50 would call this class solved. It is the worst.

Imbalance is a cost bias, not noise. At **13.8 : 1**, `door` (96 instances vs 1 341 windows) is worst on every measure. Both `door` → `window` confusions had near-perfect boxes (IoU 0.976, 0.970) — localised right, *named* wrong. Second-highest unit rate, so missing one in six understates entrance cost **systematically**.

ITERATION PLAN — DATA, NOT A BIGGER MODEL

- **150 hard negatives** of reflective opaque surfaces. The model learned "reflection ⇒ glazing" and has never seen a counterexample.
- **Rebalance `door`** 96 → ~300 instances. Targeted collection, not oversampling.
- **Small-instance coverage + an evidenced resolution decision** (640 vs 1280 vs tiling) — measured, not assumed.

A stranger can run this without asking a single question

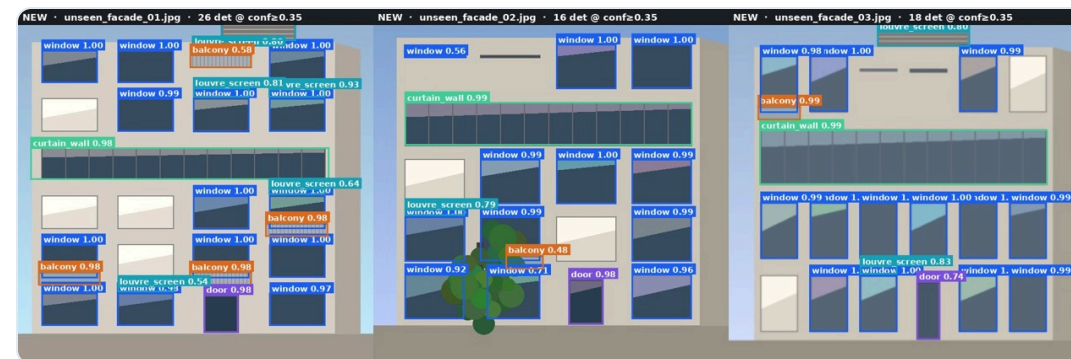
REPRODUCIBILITY PROOF

2026-09-20 19:57 UTC · restart-and-run-all from a clean checkout · **CPU only**, no GPU · **499 s** for the full 30-epoch contract · `ultralytics 8.4.157` · completed with no manual intervention.

Expected: **2–4 min** on a Colab T4, 8–12 min on CPU.

The default dataset source is **synthetic**, so the notebooks run end to end with **no account, no API key and no GPU**. That is the reproducibility requirement taken literally. Point `DATASET_SOURCE` at Roboflow for real results — the verification banner disappears automatically.

One config file. Every parameter the README quotes lives in `src/config.py` and is read from there by the notebooks. There is no second place to change them, so the documentation cannot drift from the run.



Unseen images — neither in training nor validation. "Never show a client a training image."

GOVERNANCE

Screening only. It produces False Negatives. It must NOT be the sole verifier for life-safety decisions.

Privacy by design — no `person` class, no `vehicle` class. Data minimisation applied *at the class contract*, so the model is structurally incapable of identifying anyone. No client or employer imagery.

Apache-2.0 over **AGPL-3.0** upstream — fine-tuned YOLOv8 weights inherit Ultralytics' AGPL. Commercial deployment needs compliance or an Enterprise Licence.